# MACHINE LEARNING AND PATTERN RECOGNITION

# <u>Lab Week 7</u>

Instructions:

**<u>Step 1</u>**: Import all libraries

- Numpy
- Pandas
- Matplotlib, Pyplot
- Seaborn
- SVC
- Train_test_split
- Confusion_matrix, classification_report, accuracy_score, balanced_accuracy,_score

**<u>Step 2</u>**: Load all the data from the three given csv files for ODI, T20 and test match types.

**<u>Step 3</u>**: Concatenate runs from all the match type and store in **Runs**. Similarly do for strike rate and store in **SR** and Grounds in **ground**.

**<u>Step 4</u>**: get the match type data from all csv and store in **match_type.**

***match_type = np.concatenate((np.repeat("ODI", len(odi['Runs'])), np.repeat("T20", len(t20['Runs'])), np.repeat("TEST", len(test['Runs']))))***

**<u>Step 5</u>**: Create a data frame **data_cricketer** and store Runs, SR and **match_type** in it. Perform any required data cleaning steps.

**<u>Step 6</u>**: From **data_cricketer**, strore **Runs** and **SR** in **X** and **match_type** in **Y.**

**<u>Step 7</u>**: Split the data X and Y in **x_train, x_test, y_train , y_test.** Use **test size=0.3** and **random state=0**

**<u>Step 8</u>:** Create SVC model with kernel Poly and its degree 6 (Adjust according to your model performance). Then fit the model and make prediction.

**<u>Step 9</u>**: Compute confusion matrix, accuracy and balanced accuracy (https://scikit-learn.org/stable/auto_examples/model_selection/plot_confusion_matrix.html )

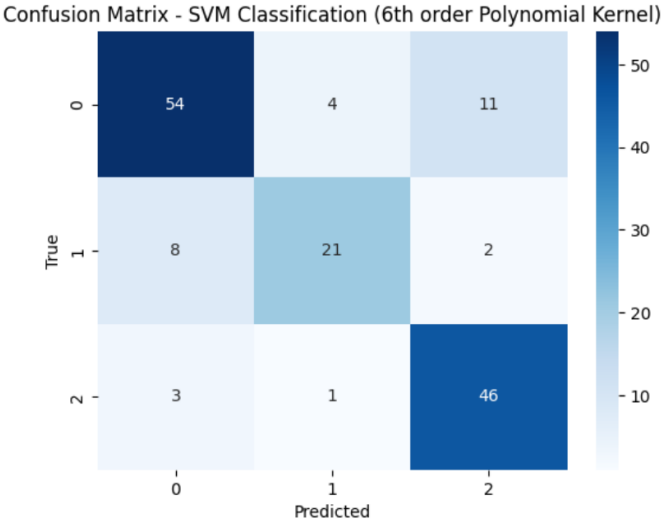
- ***confusion_matrix()***
- ***accuracy_score()***
- ***balanced_accuracy()***

***<u>Step 10</u>: Plot confusion matrix***

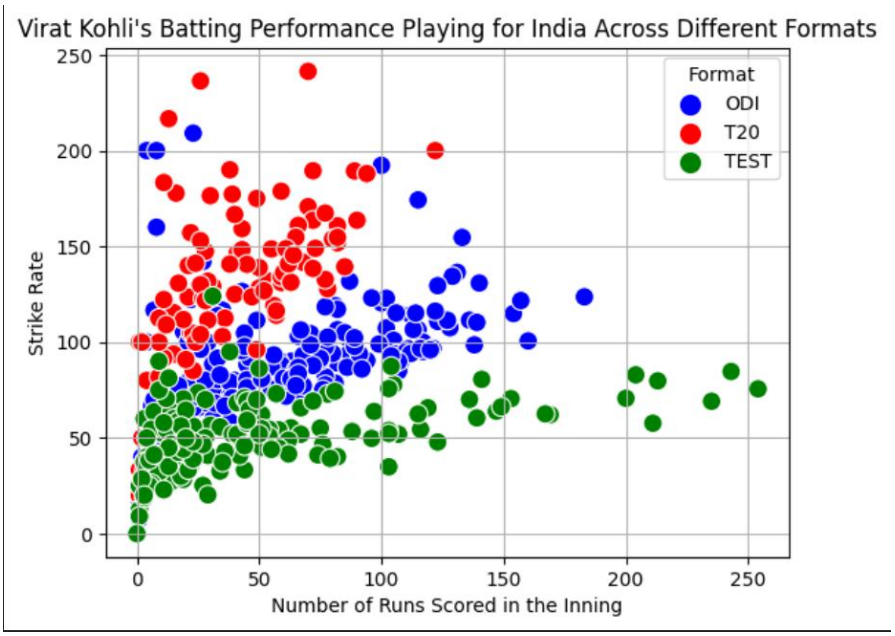
- Recommend to us ***sns.heatmap()***

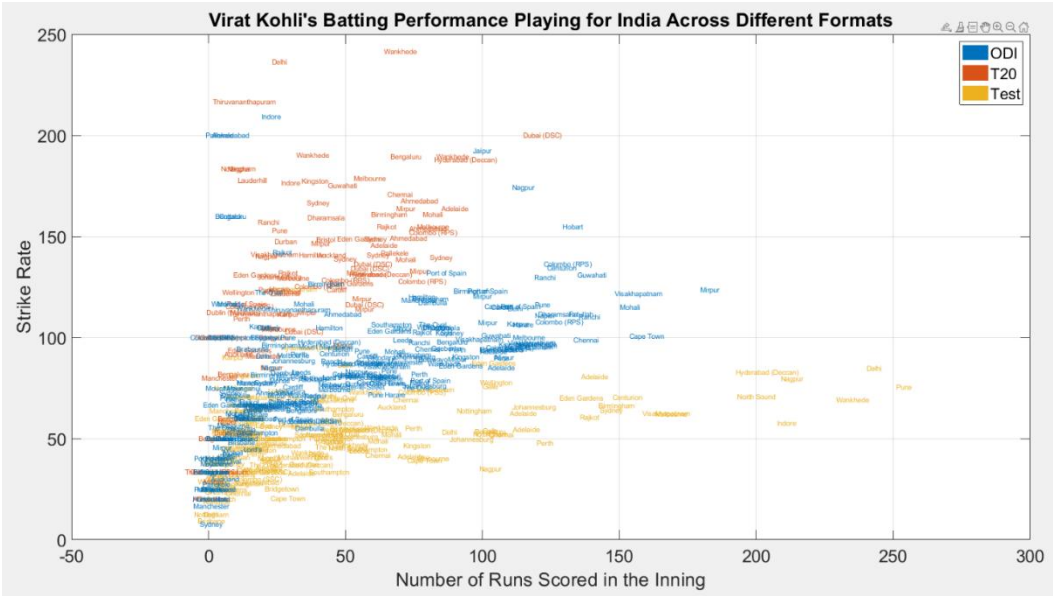
**<u>Step 11</u>:** Also print classification report that includes precision, recall, f1 score and accuracy. See the output image. (Your output may slightly differ)

- Use ***classification_report()***

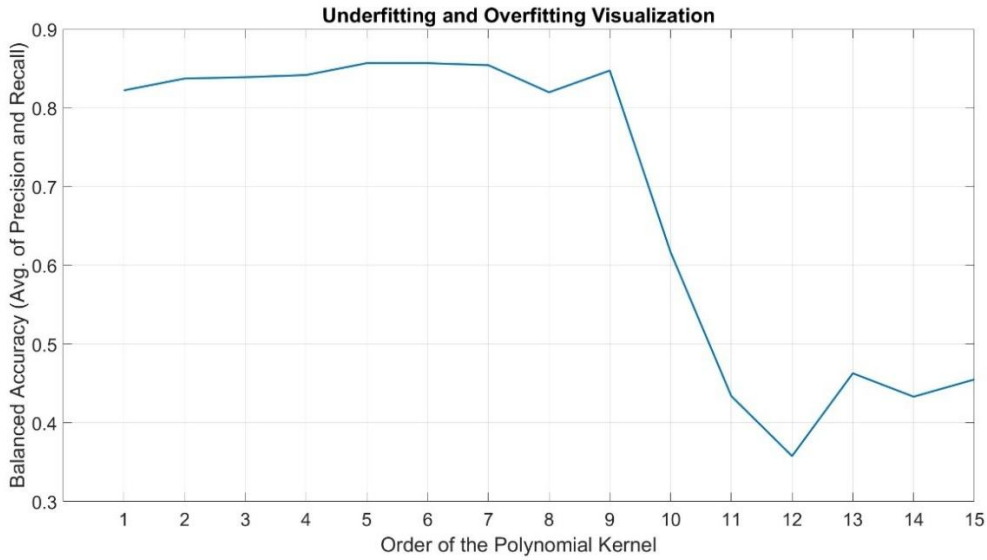



**<u>Step 12</u>**: Create a scatter plot for Runs scored on x-axis and strike rate on y-axis with match type as a grouping. See reference output plot given below (Any one of two images is fine).






Note: See why we are using polynomial degree of 6th order. If you can visualize it, plot this graph.



**<u>Report:</u> Answer the following questions within your report:**

1. What are Support Vectors in SVM? How do they help in making SVM less prone to outliers?
2. What is the difference between a hard margin and a soft margin in SVM?
3. How does SVM deal with non-linear classification boundaries?
4. What is the role of gamma in SVM? How does the value of gamma impact underfitting/overfitting?
5. How are SVMs used for multi-class classification problems?

**<u>DELIVERABLES:</u> <u>Items to be uploaded on LMS-</u>**

- Main code file along with the outputs and the report answers in one file in **PDF format.** Ensure that your file contains the following plots:

    - Scatter plot for strike rate vs runs
    - Confusion matrix as shown below in output reference image
    - Other plots shown above

- **Please note:** Write your **comments along with the code** for each step as required. All plots must be **correctly labelled**.

- Report answers must be written in **full and complete sentences**. Answer each question in a new line.

- Your file should be as follows. Yourname_lab7.pdf

- Due time and date are given on LMS. Submit it before the deadline.